Probability
Lecture 4: Continuous Random Variables

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Content

1. Continuous Random Variables
2. Expectation, Variance and Moment
3. Common Continuous Random Variables
4. Homework

Random Variables

- Recall that: $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space with Ω: set of all possible outcomes.

Definition (Precise definition)

A random variable is a measurable function $X : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, i.e. for any $a \in \mathbb{R}$,

$$\{\omega \in \Omega : X(\omega) \leq a\} \in \mathcal{F}.$$

Definition

The distribution of the discrete random variable X is given through the **distribution table**

X	x_1	x_2	$\dots$	x_n	$\dots$
$\mathbb{P}$	p_1	p_2	$\dots$	p_n	$\dots$

where $p_i = \mathbb{P}(X = x_i)$.

Cumulative Distribution Function

- Idea: the most important information of a random variable is its distribution.

Definition

Given a random variable X. Its **Cumulative Distribution Function** (CMF) is defined as

$$F_X : \mathbb{R} \to [0,1]$$
$$x \to F_X(x) = \mathbb{P}(X \leq x).$$

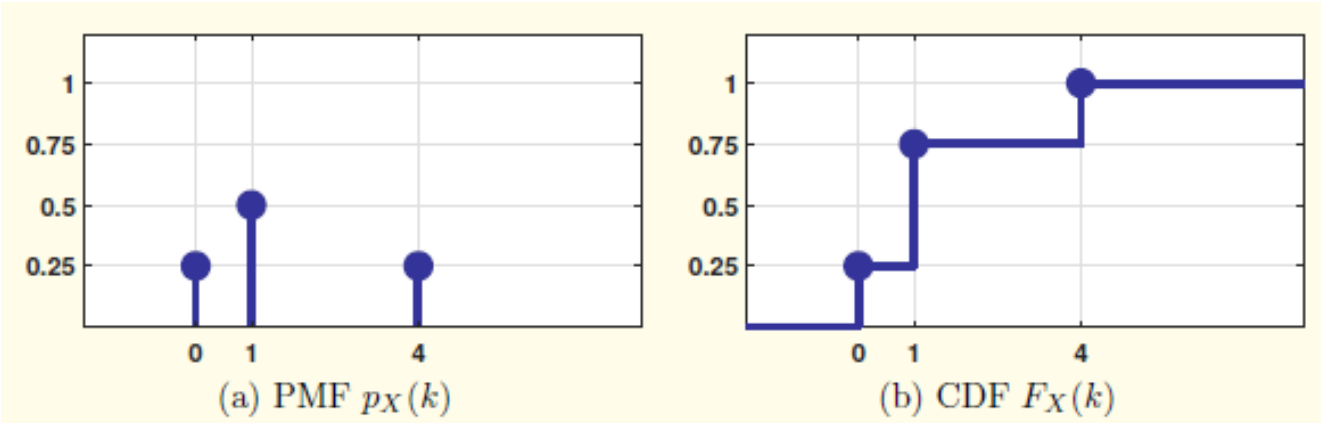


Figure: CMF of a discrete random variable has step (stairway) form.

Properties of CMF

- F_X is non-decreasing;
- F_X is right-continuous and has left limit at every point;
- $\lim_{x\to-\infty} F_X(x) = 0,\ \lim_{x\to+\infty} F_X(x) = 1.$
- For $x \leq y$ then $F_X(y) - F_X(x) = \mathbb{P}[x < X \leq y]$;

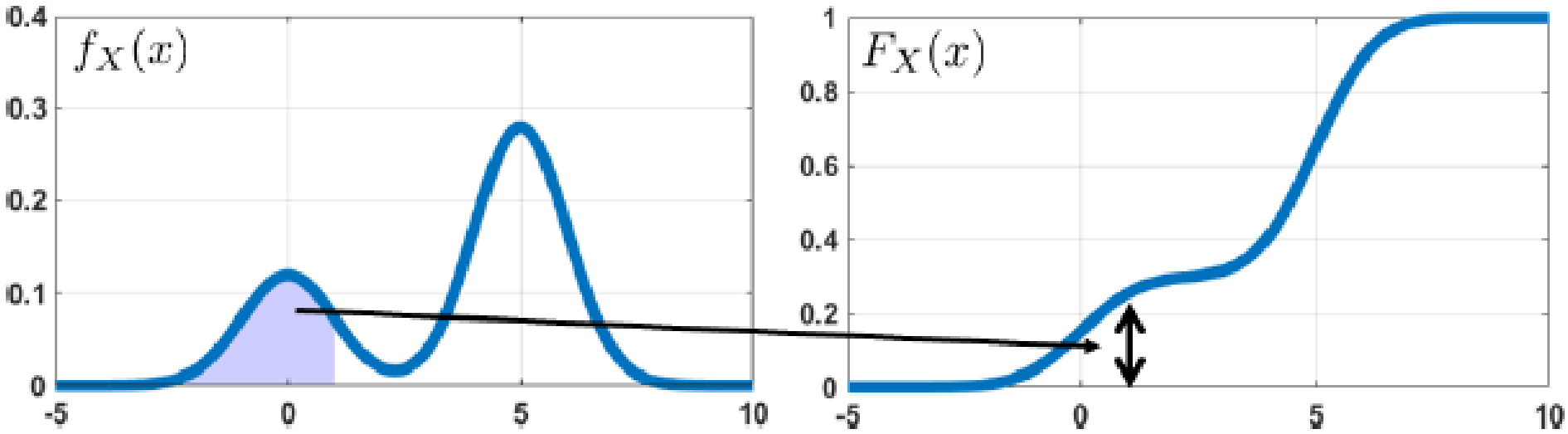
Continuous random variable

Definition

A random variable X is called **continuous** (CMF) if its cumulative distribution function is differentiable.
Define

$$f_X(x) = F'_X(x),$$

then $f_X(x)$ is called the **density function**.



Properties of density function

- $f(t)$ is non-negative: $f(t) \geq \ \forall t \in \mathbb{R}$;
- For any subset $A \in \mathcal{B}(\mathbb{R})$;

$$\mathbb{P}(X \in A) = \int_A f(t)dt.$$

+ For $A = (-\infty, x]$,

$$F_X(x) = \mathbb{P}(X \in (-\infty, x]) = \int_{-\infty}^{x} f(t)dt,$$

+ For $A = \mathbb{R}$,

$$\int_{-\infty}^{\infty} f(t)dt = 1,$$

+ For $A = [a, b]$,

$$\mathbb{P}(X \in (a, b]) = \int_a^b f(t)dt,$$

+ For $A = \{a\}]$,

$$\mathbb{P}(X = a) = \int_a^a f(t)dt = 0.$$

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Examples

- Example: Consider a random variable with density function

$$f(x) = \begin{cases} c.x & \text{for } x \in [0,1] \\ 0 & \text{otherwise} \end{cases}$$

- Find the constant c.
- Calculate the probability $\mathbb{P}(X \in [0, 0.5])$.
- Calculate the probability $\mathbb{P}(X \in (-\infty, 0.75])$.

Examples

- Example: Consider a random variable with density function

$$f(x) = \begin{cases} c.x & \text{for } x \in [0,1] \\ 0 & \text{otherwise} \end{cases}$$

- $c = 2$.

$$1 = \int_{\mathbb{R}} f(x)dx = \int_0^1 cxdx = c/2.$$

- $\mathbb{P}(X \in [0, 0.5]) = \int_0^{1/2} f(x)dx = \int_0^1 2xdx = 1/4.$
- $\mathbb{P}(X \in [0, 0.75]) = \int_{-\infty}^{3/4} f(x)dx = \int_0^{3/4} 2xdx = 9/16.$

Examples

- Example: Consider a random variable with density function

$$f(x) = \begin{cases} c.x.(1-x) & \text{for } x \in [0,1] \\ 0 & \text{otherwise} \end{cases}$$

- Find the constant c.
- Calculate the probability $\mathbb{P}(X \in [0, 0.5])$.

1 Continuous Random Variables

2 Expectation, Variance and Moment

3 Common Continuous Random Variables

4 Homework

Expectation

- Recall that: Given a discrete random variable X with distribution

X	x_1	x_2	$\dots$	x_n	$\dots$
$\mathbb{P}$	p_1	p_2	$\dots$	p_n	$\dots$

Then the expectation of X is

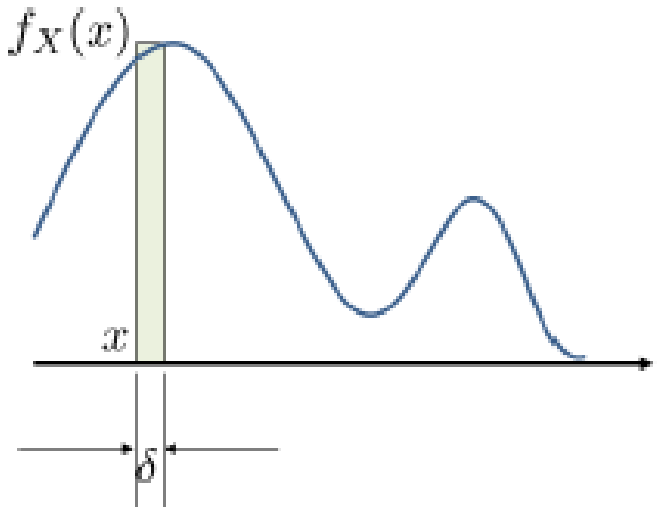
$$\mathbb{E}(X) = \sum_{x_i \in X(\Omega)} x_i p_i.$$

- Meaning: Expectation = Average Value= Mean.

Expectation of a continuous random variable

- Approximate a continuous random variable by discrete ones. In a small interval $X \in [x_i, x_i + \delta]$, approximate $X(\omega)$ by $X_\delta(\omega) = x_i$.

$$\mathbb{P}[X \in [x, x+\delta]] = \int_x^{x+\delta} f(t)dt \approx \delta f(x).$$



- We have

$$\mathbb{E}[X_\delta] = \sum_i x_i.\mathbb{P}(X_\delta = x_i) = \sum_i x_i.f(x_i).\delta.$$

Expectation, Moment and Variance

Theorem

Given a discrete random variable X with density function $f(x)$. Then its expectation is defined as

$$\mathbb{E}[X] = \int_{\Omega} X d\mathbb{P} = \int_{\mathbb{R}} x.f(x)d(x).$$

Expectation, Moment and Variance

Theorem

In general,

- *For any function* $g: \mathbb{R} \to \mathbb{R}$,

$$\mathbb{E}[g(X)] = \int_R g(x)f(x)dx.$$

- *Moment: for any* k

$$\mathbb{E}[X^k] = \int_R x^k f(x)dx.$$

- *Variance*

$$Var(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

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Theorem

In general,

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Example

- Example: Consider a random variable with density function

$$f(x) = \begin{cases} 2.x & \text{for } x \in [0,1] \\ 0 & \text{otherwise} \end{cases}$$

- Find the expectation $\mathbb{E}[X]$.
- Calculate the moments $\mathbb{E}[X^k]$ for $k \geq 0$.
- Calculate the variance.
- Let $Y = \sin(\pi X)$. Find $\mathbb{E}[Y]$ and $\mathbb{E}[Y^2]$.

- Find the expectation $\mathbb{E}[X]$.

$$\mathbb{E}[X] = \int_{\mathbb{R}} xf(x)dx = \int_0^1 x.2xdx = \int_0^1 2x^2dx = 2/3.$$

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$$\mathbb{E}[X^k] = \int_{\mathbb{R}} x^k f(x)dx = \int_0^1 x^k.2xdx = \int_0^1 2x^{k+1}dx = \frac{2}{k+2}.$$

- Calculate the variance.

$$Var(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2 = \frac{2}{4} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}.$$

- Let $Y = \sin(\pi X)$. Find $\mathbb{E}[Y]$ and $\mathbb{E}[Y^2]$.

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$$E[Y] = \int_{\mathbb{R}} \sin(\pi x) f(x) dx = \int_0^1 \sin(\pi x).2x dx = 2 \int_0^1 x d\left(\frac{-\cos(\pi x)}{\pi}\right)$$

$$= 2\left[x.\frac{-\cos(\pi x)}{\pi}\right]\Big|_0^1 + 2\int_0^1 \frac{\cos(\pi x)}{\pi} dx = \frac{2}{\pi}.$$

$$\mathbb{E}[Y^2] = \int_{\mathbb{R}} \sin^2(\pi x) f(x) dx = \int_0^1 \sin^2(\pi x).2x dx = \frac{1}{2}.$$

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Example

- Example: Consider a random variable with density function

$$f(x) = \begin{cases} 6.x.(1-x) & \text{for } x \in [0,1] \\ 0 & \text{otherwise} \end{cases}$$

- Find the expectation $\mathbb{E}[X]$.
- Calculate the variance.

Integrable Random Variable

- Example: Let X be a **Cauchy random variable**, i.e, its density function is

$$f(x) = \frac{1}{\pi(1+x^2)} \quad \forall x \in \mathbb{R}.$$

+ Naive calculation

$$\mathbb{E}[X] = \int_R \frac{x}{\pi(1+x^2)} dx = \int_{-\infty}^{0} + \int_0^{\infty},$$

$\Rightarrow$ Undefined.

Definition

A random variable X is called **absolutely integrable** if

$$\mathbb{E}(|X|) = \int_\Omega |X| d\mathbb{P} = \int_{\mathbb{R}} |x| f(x) dx < \infty.$$

We say that X has an expectation if it is absolutely integrable.

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Properties of Variance

Theorem

The variance of a random variable X has the following properties:

(i). **Moment**.

$$Var(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

(ii). **Scale**. *For any constant c,*

$$Var(cX) = c^2\, Var(X).$$

(iii). **DC Shift**. *For any constant c,*

$$Var(X + c) = Var(X).$$

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4 Homework

Common Continuous Random Variables

- Uniform ($U([a, b])$).
- Exponential ($\text{Exp}(\lambda)$).
- Gaussian ($\mathcal{N}(\mu, \sigma^2)$). Normal distribution.

Uniform distribution

Definition

We say that X is drawn from a uniform distribution on an interval $[a, b]$ if its density function is

$$f(x) = \begin{cases} \dfrac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise.} \end{cases}$$

We write $X \sim U[a, b)$.

Theorem

If $X \sim U[a, b)$ then

$$\mathbb{E}[X] = \frac{a+b}{2}, \quad Var(X) = \frac{(b-a)^2}{12}.$$

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Exponential Random Variable

Definition

We say that X is drawn from an exponential distribution of parameter $\lambda > 0$ if its density function is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

We write $X \sim Exp(\lambda)$.

Theorem

If $X \sim Exp(\lambda)$ then

$$F_X(x) = \begin{cases} 0 & for\ x < 0 \\ 1 - \lambda e^{-\lambda x} & x \geq 0, \end{cases}$$

$$\mathbb{E}[X] = 1/\lambda, \quad Var(X) = 1/\lambda^2.$$

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Exponential Random Variable

- Recall that $X \sim Poi(\lambda)$ if

$$\mathbb{P}(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}.$$

$\rightarrow$ number of occurrences in a period of time.

- Let's consider the probability of the **waiting time** for the first (next) occurrence.

+ During t units time, the average number of occurrences is λt .

+ Model: $X_t \sim Poi(\lambda t)$.

+ Waiting time is bigger than t means that NO occurrence during t units time

$$\mathbb{P}(W > t) = \mathbb{P}(X_t = 0) = e^{-\lambda t}\frac{(\lambda.t)^0}{0!} = e^{-\lambda t}.$$

- Other models: for life time.

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Gaussian Random Variables

Definition

We say that X is drawn from a Gaussian (or Normal) distribution of parameter (μ, σ^2) if its density function is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)$$

We write $X \sim \mathit{Gaussian}(\mu, \sigma^2)$ or $X \sim \mathcal{N}(\mu, \sigma^2)$.

Definition

When $\mu = 0$ and $\sigma^2 = 1$, we have $X \sim \mathcal{N}(0, 1)$ with density function

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-x^2}{2}\right).$$

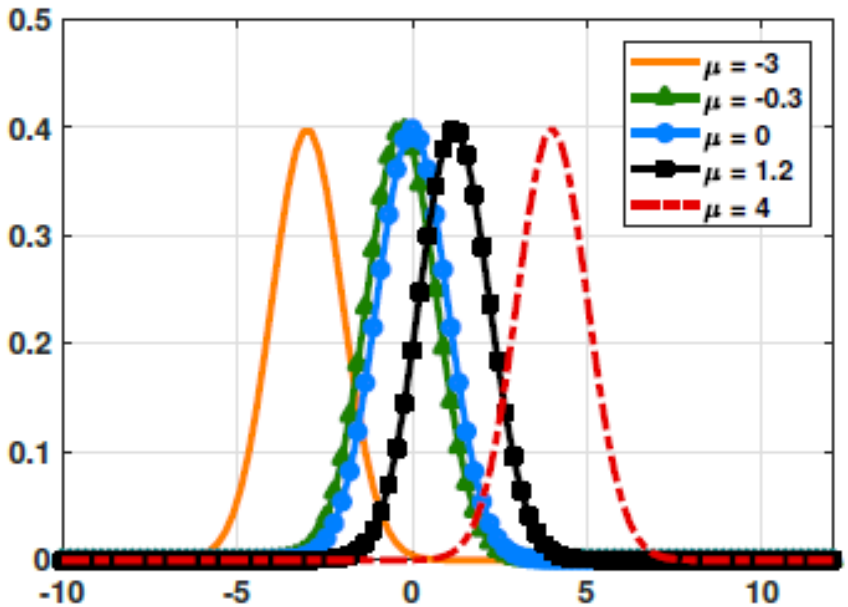
It is called **standard normal distribution**.

$\rightarrow$ Gaussian integral identity!

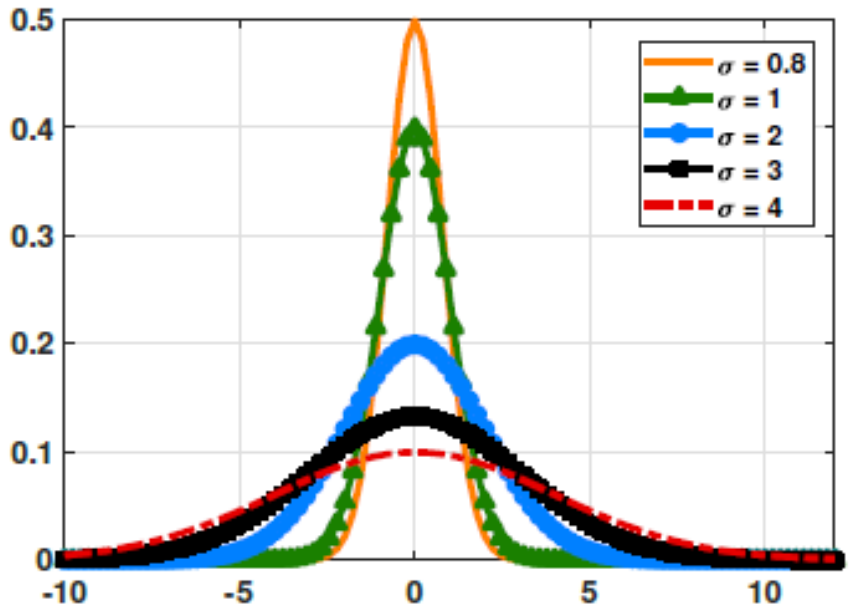
Gaussian Random Variables

$\rightarrow$ Gaussian integral identity!

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-x^2/2} dx = 1.$$



μ changes, $\sigma = 1$



$\mu = 0$, σ changes

Gaussian Random Variables

Theorem

If $X \sim \mathcal{N}(\mu, \sigma^2)$ *then*

$$\mathbb{E}[X] = \mu; \quad Var(X) = \sigma^2.$$

$$E[X] = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{\mathbb{R}} x \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right) dx$$

$$\overset{(x=\mu+\sigma y)}{=} \frac{1}{\sqrt{2\pi\sigma^2}} \int_{\mathbb{R}} (\mu + \sigma y) \exp\left(-y^2/2\right) \sigma dy$$

$$= \frac{\mu}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp\left(-y^2/2\right) dy + \frac{\sigma}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp y \left(-y^2/2\right) dy$$

$$= \mu + 0 = \mu.$$

Gaussian Random Variables

Theorem

If $X \sim \mathcal{N}(\mu, \sigma^2)$ *then*

$$\mathbb{E}[X] = \mu; \quad Var(X) = \sigma^2.$$

$$\begin{aligned} E[X] &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{\mathbb{R}} x \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right) dx \\ &\overset{(x=\mu+\sigma y)}{=} \frac{1}{\sqrt{2\pi\sigma^2}} \int_{\mathbb{R}} (\mu + \sigma y) \exp\left(-y^2/2\right) \sigma dy \\ &= \frac{\mu}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp\left(-y^2/2\right) dy + \frac{\sigma}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp y \left(-y^2/2\right) dy \\ &= \mu + 0 = \mu. \end{aligned}$$

Gaussian Random Variables

$$
\begin{aligned}
\mathrm{Var}(X) &= \mathbb{E}[(X - \mathbb{E}X)^2] \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{\mathbb{R}} (x-\mu)^2 \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right) dx \\
&\overset{(x=\mu+\sigma y)}{=} \frac{1}{\sqrt{2\pi\sigma^2}} \int_{\mathbb{R}} \sigma^2 y^2 \exp\left(-y^2/2\right) \sigma dy \\
&= \frac{\sigma^2}{\sqrt{2\pi}} \int_{\mathbb{R}} y^2 \exp\left(-y^2/2\right) dy \\
&= \frac{\sigma^2}{\sqrt{2\pi}} \left[\left(-ye^{-y^2/2}\right)\Big|_{-\infty}^{\infty} + \int_{\mathbb{R}} \exp\left(-y^2/2\right) dy \right] = \sigma^2,
\end{aligned}
$$

$(u = -y \text{ and } v = e^{-y^2/2})$.

Gaussian Random Variables

Theorem

If $X \sim \mathcal{N}(\mu, \sigma^2)$ *then*

$$Y = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1).$$

Conversely is also true.

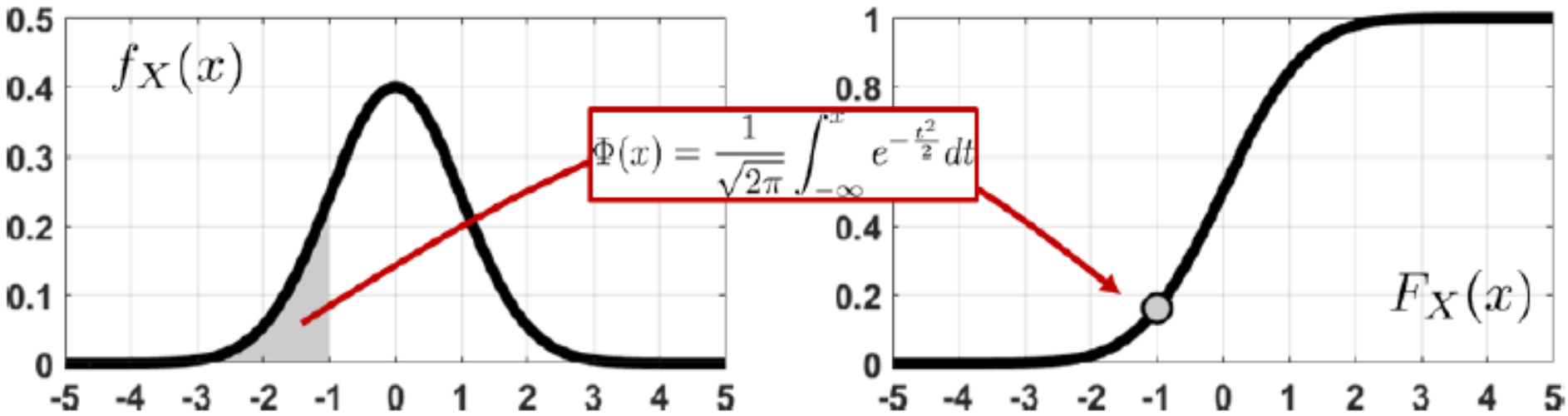
$$\begin{aligned}
\mathbb{P}(Y \leq t) &= \mathbb{P}\left(\frac{X - \mu}{\sigma} \leq t\right) = \mathbb{P}(X \leq \mu + \sigma t) \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\mu+\sigma t} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right) dx \\
&\overset{(x=\mu+\sigma y)}{=} \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{t} \exp\left(-y^2/2\right) \sigma dy \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{t} \exp\left(-y^2/2\right) dy
\end{aligned}$$

Gaussian Random Variabless

Definition

The Cumulative Distribution Function of a standard normal random variable is denoted by $\Phi(.)$

$$\Phi(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{t} \exp\left(-x^2/2\right) dx.$$



Gaussian Random Variables

Theorem

If $X \sim \mathcal{N}(\mu, \sigma^2)$ *then*

- $F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}\left(\frac{X-\mu}{\sigma} \leq \frac{x-\mu}{\sigma}\right) = \Phi\left(\frac{x-\mu}{\sigma}\right).$
- $\mathbb{P}(X \geq x) = 1 - F_X(x) = 1 - \Phi\left(\frac{x-\mu}{\sigma}\right).$
- $\mathbb{P}(X \in [a, b]) = \Phi\left(\frac{xb-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right).$

- Example: the distribution of the IQ scores follows a normal distribution with mean $\mu = 100$ and standard deviation $\sigma = 15$.
+ Find the probability that a random person has IQ score at most 85.
+ Find the probability that a random person has IQ score at least 111.
+ To be a MENSA member, you have to get your IQ score bigger than at least 98% of world population. What is the possible smallest score to be in MENSA?

Gaussian Random Variables

Theorem

If $X \sim \mathcal{N}(\mu, \sigma^2)$ *then*

- $F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}\left(\frac{X-\mu}{\sigma} \leq \frac{x-\mu}{\sigma}\right) = \Phi\left(\frac{x-\mu}{\sigma}\right).$
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Gaussian Random Variables

- Example: the distribution of the IQ scores follows a normal distribution with mean $\mu = 100$ and standard deviation $\sigma = 15$.

+ Find the probability that a random person has IQ score at most 85. $z = -1$, 16%

+ Find the probability that a random person has IQ score at least 111.$z = 0.73$, 23%

+ MENSA member IQ at least 130.8.

- Terence Tao IQ Score: 230-240
- Kim Ung-Yong IQ Score: 210 PhD at 15yrs
- Garry Kasparov IQ Score: 190

Functions of Random Variables

- Given a random variable X. For a function $g : \mathbb{R} \to \mathbb{R}$, define $Y = g(X)$.

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y).$$

- If g is monotonically increasing, then

$$F_Y(y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \leq g^{-1}(y)) = F_X(g^{-1}(y)).$$

+ X has density function $f_X(x)$

$$F_Y(y) = \int_{-\infty}^{g^{-1}(y)} f_X(x)dx.$$

+ Hence Y has density function $f_Y(t)$

$$f_Y(y) = \frac{d}{dy}F_Y(y) = \frac{d}{dy}\int_{-\infty}^{g^{-1}(y)} f_X(x)dx = f_X(g^{-1}(y))\frac{dg^{-1}(y)}{dy}.$$

Example

- Let X be a random variable with density function

$$f_X(x) = ae^x e^{-ae^x}.$$

Let $Y = e^X$, and find $f_Y(y)$.
+ Hint: $g(x) = e^x$ then $g^{-1}(y) = \log y$.

$$f_Y(y) = f_X(g^{-1}(y))\frac{dg^{-1}(y)}{dy} = ae^{\log y}e^{-ae^{\log y}}(\log y)' = ae^{-ay}.$$

$$\Rightarrow Y \sim Exp(a).$$

Example

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$\Rightarrow Y \sim Exp(a)$.

Homework

+ Stanley Chan: Chapter4. 1-15.
+ Ngo Hoang Long: Chapter2. 30, 32, 33, 35, 36.

Thank you!